Donation Notification Automation

Technical Design Document (TDD)

Salesforce + Slack Integration

Project Title	Donation Notification Automation — Salesforce + Slack Integration
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1. Introduction

1.1 Purpose

This Technical Design Document (TDD) is derived from the Salesforce + Slack Integration TID v1.0. It defines the detailed design specifications, component logic, and test case matrix for the Donation Notification Automation. The TDD serves as the execution reference for developers, admins, and QA teams.

1.2 Scope

This TDD covers technical design for the following implementation areas:

- Integration Foundation — Salesforce–Slack OAuth connection and notification toggle
- Security & Permissions — Slack User and Slack Channel Manager permission sets; user account linking
- Slack Channel Architecture — three dedicated notification channels with Salesforce bot integration
- Automation (Flow Builder) — three Record-Triggered Flows on Donation__c for new donation, stage change, and high-value alerts
- Troubleshooting — root cause and resolution design for all known failure scenarios

2. System Prerequisites

2.1 Environment Requirements

The following conditions must be met before any implementation step begins:

Prerequisite	Required Specification	Verification Method
Salesforce Edition	Enterprise, Unlimited, or Developer Edition	Setup > Company Information > Edition
User Authority	Salesforce System Administrator profile	Setup > Users > confirm profile = System Administrator
Slack Authority	Slack Workspace Admin or App install permission	Slack > Settings & Permissions > verify admin rights
Browser	Chrome or Firefox (required for OAuth popup stability)	Safari not supported — OAuth popup may fail
Donation__c — Amount__c	Currency field must exist on Donation__c	Setup > Object Manager > Donation__c > Fields
Donation__c — Stage__c	Picklist field must exist on Donation__c	Setup > Object Manager > Donation__c > Fields
Donation__c — Donor_Name__c	Text or Lookup field must exist on Donation__c	Setup > Object Manager > Donation__c > Fields
Donation__c — Donation_Date__c	Date field must exist on Donation__c	Setup > Object Manager > Donation__c > Fields

3. Phase 1 — Integration Foundation

3.1 Overview

This phase establishes the trusted OAuth connection between Salesforce and the Slack workspace. All three notification flows depend on this connection — without a confirmed 'Connected' status, no Slack messages can be posted by any flow.

3.2 Execution Steps

Step	Action	Navigation	Expected Outcome
1	Open Salesforce Setup	Gear icon (top-right) > Setup	Setup console opens
2	Search for Slack Setup	Quick Find bar > type 'Slack Setup' > click result	Slack Setup wizard opens in main panel
3	Initiate OAuth connection	Click 'Connect to Slack'	OAuth popup opens in browser; workspace selection screen appears
4	Authorise the workspace	Select correct workspace > click 'Allow' in popup	Popup closes; page redirects to Slack Setup
5	Verify Connected status	Inspect status badge on Slack Setup page	Green 'Connected' badge displayed next to workspace name
6	Enable notification toggle	Toggle 'Allow Salesforce to Send Notifications to Slack' to ON	Toggle active; Salesforce bot appears in Slack under Apps

3.3 Design Notes

- The notification toggle must be enabled only after the 'Connected' badge is confirmed — enabling before connection has no effect
- If the OAuth popup is blocked, the admin must allow popups for the Salesforce domain in the browser settings
- A successful OAuth redirect without the green badge indicates a partial failure — reconnect must be performed

4. Phase 2 — Security & Permissions

4.1 Permission Set Architecture

Two Salesforce Permission Sets control access to Slack integration features. Both must be assigned correctly for full notification functionality.

Permission Set	Functional Capability	Assignment Target
Slack User	Allows users to link their personal Slack account to Salesforce for individual notification receipt. Without this, the user cannot receive any Slack alerts.	All Salesforce users who need to receive Slack donation notifications
Slack Channel Manager	Allows admin-level management of Slack channel connections within Salesforce Setup.	Salesforce System Administrators only — not for standard or community users

4.2 Permission Set Assignment Steps

Individual Assignment

1. Setup > Users > select target user
2. Scroll to Permission Set Assignments > click Edit Assignments
3. Move 'Slack User' to Enabled column > Save
4. For admin users: also move 'Slack Channel Manager' to Enabled > Save

Bulk Assignment

5. Setup > Permission Sets > select 'Slack User'
6. Click Manage Assignments > Add Assignments
7. Select all required users > click Assign
8. Verify all selected users appear in the Assigned Users list

4.3 User Account Linking Protocol

After permission set assignment, each user must perform a one-time Slack account link. This step cannot be completed by an admin on behalf of a user.

Step	User Action	Navigation	Expected Outcome
1	Open personal profile settings	User avatar (top-right) > Settings	Settings panel opens
2	Locate Slack connection	Settings > Slack OR Settings > Connected Accounts	Slack connection option visible
3	Connect Slack account	Click 'Connect to Slack'	OAuth popup opens for personal account authorization
4	Authorise	Select personal Slack workspace > click Allow	Popup closes; settings page reloads
5	Confirm status	Inspect connection status on Settings page	Status displays 'Connected' — user will now receive Slack notifications

5. Phase 3 — Slack Channel Architecture

5.1 Channel Design Strategy

Three dedicated Slack channels route notifications to the correct audience by event type. This tiered approach prevents notification fatigue — each team receives only the alerts relevant to their role.

Channel Name	Triggering Event	Target Audience	Purpose
#new-donations	Creation of any Donation__c record	Fundraising & Development	Real-time visibility of all new donations
#donation-stage-updates	Change to the Stage__c picklist field	Ops & Donor Relations	Track donation lifecycle stage transitions
#high-value-donations	Amount__c >= 100,000	Executive Leadership	Immediate alert for major gifts

5.2 Channel Creation & Bot Integration

Naming Convention

All channel names must use lowercase letters and hyphens only — no spaces, uppercase letters, or special characters. The '#' prefix is added automatically by Slack and must NOT be typed in the name field.

Channel	Correct Name (enter in Slack)	Common Mistake to Avoid
New Donations	new-donations	Do NOT enter: '#new-donations' or 'New Donations' or 'new_donations'
Stage Updates	donation-stage-updates	Do NOT enter: '#donation-stage-updates' or spaces
High Value Donations	high-value-donations	Do NOT enter: '#high-value-donations' or 'high value'

Salesforce App Integration — Steps per Channel

The Salesforce App bot must be manually added to each channel. Without this step, flows cannot post messages to the channel even if the flow is active and the connection is confirmed.

9. Open the channel in Slack
10. Click the channel name at the top > View channel details
11. Navigate to Integrations tab > click Add Apps
12. Search for 'Salesforce' > click Add
13. Verify '@Salesforce' bot appears in the channel member list
14. Repeat all steps for each of the three channels

6. Phase 4 — Automation (Flow Builder)

6.1 Object Schema Requirements

All three flows are built on the Donation__c custom object. The following fields must be confirmed as deployed before any flow is created or activated:

Field	Data Type	Role in Flows
Amount__c	Currency	Flow 3 condition field — Amount__c >= 100,000 determines high-value threshold
Stage__c	Picklist	Flow 2 condition field — ISCHANGED(Stage__c) drives the stage-change entry condition
Donor_Name__c	Text/Lookup	Used in Slack message content across all 3 flows to identify the donor
Donation_Date__c	Date	Available for message content enrichment in all flows

6.2 Flow 1 — New Donation Alert

Parameter	Configuration Detail
Flow Label	New Donation — Slack Alert
Flow Type	Record-Triggered Flow
Object	Donation__c
Trigger Event	A record is Created
Entry Condition	None — fires on every new Donation__c record regardless of amount or stage
Slack Channel	#new-donations — enter in Flow as 'new-donations' (no '#' prefix)
Message Content	New Donation Received: {Donor_Name__c} — Amount: {Amount__c}
Optimization	Actions and Related Records
Error Handling	Flow Fault Path configured — failure sends alert email to System Admin; record save is not blocked

6.3 Flow 2 — Stage Update Alert

Parameter	Configuration Detail
Flow Label	Donation Stage Change — Slack Alert
Flow Type	Record-Triggered Flow
Object	Donation__c
Trigger Event	A record is Updated
Entry Condition	ISCHANGED({!$Record.Stage__c}) = True — fires only when Stage__c field value actually changes
Slack Channel	#donation-stage-updates — enter as 'donation-stage-updates' (no '#' prefix)
Message Content	Donation Stage Updated: {Donor_Name__c} — New Stage: {Stage__c}

Optimization	Actions and Related Records
Error Handling	Flow Fault Path configured — failure logged; record update is not blocked

ISCHANGED() Logic

The ISCHANGED() condition prevents false alerts when non-stage fields (such as Description or Amount) are edited on an existing Donation record without any Stage change.

Edit Action	ISCHANGED(Stage__c)	Flow 2 Fires?
Stage changed from Pledged to Received	True	YES — message posted to #donation-stage-updates
Description field edited; Stage unchanged	False	NO — entry condition not met; no alert sent
Amount updated; Stage unchanged	False	NO — ISCHANGED only monitors Stage__c field
Stage changed from Received to Failed	True	YES — message posted to #donation-stage-updates

6.4 Flow 3 — High-Value Alert

Parameter	Configuration Detail
Flow Label	High Value Donation — Slack Alert
Flow Type	Record-Triggered Flow
Object	Donation__c
Trigger Event	A record is Created or Updated
Entry Condition	Amount__c Greater Than or Equal To 100,000
Slack Channel	#high-value-donations — enter as 'high-value-donations' (no '#' prefix)
Message Content	HIGH VALUE DONATION: {Donor_Name__c} — Amount: {Amount__c} — Stage: {Stage__c}
Optimization	Actions and Related Records
Error Handling	Flow Fault Path configured — failure sends priority alert to System Admin

Created OR Updated Trigger — Rationale

- Created trigger: catches new donations entered directly at or above the 100,000 threshold
- Updated trigger: catches donations that begin below the threshold and are later amended to a high-value amount
- Both Flow 1 and Flow 3 fire independently when a new high-value donation is created — this dual-channel behaviour is intentional

6.5 Flow Interaction Summary

Scenario	Flow 1 Fires?	Flow 3 Fires?	Channels Notified
New donation — Amount = 500	YES	NO	#new-donations only
New donation — Amount = 150,000	YES	YES	#new-donations AND #high-value-donations
Stage change on existing donation	NO	NO	#donation-stage-updates only (Flow 2)
Amount updated from 500 to 150,000	NO	YES	#high-value-donations only
Description edited; Stage unchanged	NO	NO	No channel — no flow fires

7. Troubleshooting Guide

7.1 Issue Resolution Reference

Symptom	Root Cause	Resolution Steps
No Slack message after Donation__c created or updated	Flow not Active, OR '#' prefix used in channel name inside the Flow	1. Flow Builder > verify Flow Status = Active 2. If Inactive, click Activate 3. Inspect the Slack action — channel name must have NO '#' prefix (use 'new-donations' not '#new-donations') 4. Edit channel name if needed > re-activate
OAuth connection error — integration fails	Salesforce–Slack OAuth connection dropped or expired	1. Setup > Slack Setup > click Reconnect 2. Re-complete OAuth popup: select workspace > Allow 3. Verify green 'Connected' badge reappears 4. Confirm notification toggle is still ON after reconnect
Duplicate Slack alerts for the same event	Multiple active Flow versions exist for the same trigger	1. Setup > Flows > search affected flow name 2. Identify all versions — only ONE should be Active 3. Deactivate all older or duplicate versions 4. Confirm only the current version remains Active
Bot not posting to channel despite active Flow	Salesforce App not added to the specific Slack channel	1. Open Slack channel > channel name > View details > Integrations 2. Add Apps > search 'Salesforce' > Add 3. Verify '@Salesforce' appears in channel member list 4. Re-test by creating a Donation__c record
User not receiving notifications	'Slack User' permission set missing, or personal Slack account not linked	1. Setup > Users > find user > verify 'Slack User' in Permission Set Assignments 2. If missing: assign 'Slack User' permission set 3. User: Profile > Settings > Slack > Connect to Slack > Authorize 4. Verify 'Connected' status in user settings